Chapter 7. Follow the needles that point north: framed cobordism and the Pontryagin construction

Milnor, "Topology from the Differentiable Viewpoint", §7, rewritten for a first-year student who wants to know what the theorem is for. Every proof is complete; one geometric step (the product neighbourhood of a cobordism near its ends) is spelled out where Milnor writes "completely analogous".

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0. A magnet full of compass needles

Inside a magnet, every point carries a little arrow: the direction of the local magnetization, a unit vector $n(x) \in S^2$. Far from any defects the arrows vary smoothly. Now mark the points where the arrow points *exactly north*. In a three-dimensional sample this is a curve (three equations in three unknowns minus… no: two equations, since a direction has two degrees of freedom, so a one-dimensional set). Along that curve, also record which way the arrow tilts as you step off the curve: eastward in one direction, upward in another. This decorated curve is the *Pontryagin manifold* of the texture.

The theorem of this chapter says that the decorated curve knows everything about the texture that survives continuous deformation. Two magnetization patterns can be deformed into one another if and only if their decorated "north curves" can be deformed into one another, through decorated surfaces in space-time. And every decorated curve arises from some texture. A question about maps to the sphere, which are hard to draw, becomes a question about curves in space, which are easy to draw. When the north curve is a knot or a link, the texture is a *hopfion*, and the linking of the north curve with the south curve is an integer the texture cannot lose. Such textures were photographed in a magnet in 2023.

Mathematically: the degree of Chapter 5 was defined for maps between manifolds of the *same* dimension. Pontryagin's construction handles maps $f : M \to S^p$ from a compact manifold of any dimension $m \geq p$ to a sphere, by replacing the finite set $f^{-1}(y)$ and its signs with the $(m-p)$-manifold $f^{-1}(y)$ and a *framing* of its normal directions. When $m = p$ we recover the degree, and as a corollary we get Hopf's theorem: maps $M^m \to S^m$ are classified up to homotopy by their degree alone.

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1. Cobordism and framings (Milnor pp. 41–42)

Throughout, M is a compact smooth manifold of dimension m without boundary, $M \subseteq \mathbb{R}^k$.

Definition. Let $N, N' \subseteq M$ be compact submanifolds of dimension n without boundary. Their *codimension* is $m - n$. N is *cobordant to N' within M* if there is a compact manifold with boundary $X \subseteq M \times [0,1]$ such that

$$X \cap (M \times [0,\varepsilon)) = N \times [0,\varepsilon), \qquad X \cap (M \times (1-\varepsilon,1]) = N' \times (1-\varepsilon,1]$$

for some $\varepsilon > 0$, $\partial X = N \times \{0\} \cup N' \times \{1\}$, and X meets $M \times \{0,1\}$ only in ∂X. Cobordism is an equivalence relation: reflexive via $N \times [0,1]$, symmetric via $t \mapsto 1 - t$, transitive by stacking two cobordisms end to end (the product structure near the ends makes the union smooth).

Definition. A *framing* of a submanifold $N \subseteq M$ of codimension p is a smooth function v assigning to each $x \in N$ a basis $v(x) = (v^1(x), \dots, v^p(x))$ of the normal space $TN_x^\perp \subseteq TM_x$. The pair (N, v) is a *framed submanifold*. Two framed submanifolds (N, v), (N', w) are *framed cobordant* if there is a cobordism X between them with a framing u of X in $M \times [0,1]$ such that

$$u^i(x,t) = (v^i(x), 0) \text{ for } (x,t) \in N \times [0,\varepsilon), \qquad u^i(x,t) = (w^i(x), 0) \text{ for } (x,t) \in N' \times (1-\varepsilon, 1].$$

This too is an equivalence relation, for the same reasons.

Not every submanifold admits a framing: a normal bundle can be twisted (the centre circle of a Möbius band inside the band). A framed submanifold is in particular one whose normal bundle is trivial.

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2. The Pontryagin manifold of a map (Milnor pp. 42–45)

Let $f : M \to S^p$ be smooth and $y \in S^p$ a regular value. By Chapter 2, Lemma 2.1, $N := f^{-1}(y)$ is a compact submanifold of codimension p, and by Lemma 2.2 its tangent space at x is $\ker df_x$. Hence df_x restricted to the normal space $TN_x^\perp$ is injective, and, both spaces having dimension p, an isomorphism onto $T(S^p)_y$. Choose a positively oriented basis $v = (v_1, \dots, v_p)$ of $T(S^p)_y$. For each $x \in N$ let $w^i(x) \in TN_x^\perp$ be the unique normal vector with $df_x(w^i(x)) = v_i$. The w^i are smooth in x (they solve a linear system whose coefficients depend smoothly on x and whose matrix is invertible), and form a framing $w = f^*v$ of N.

Definition. $(f^{-1}(y), f^*v)$ is the *Pontryagin manifold* of f (at y, with respect to v).

Theorem A. The framed cobordism class of the Pontryagin manifold does not depend on the choice of regular value y or of positive basis v.

Theorem B. Two smooth maps $M \to S^p$ are smoothly homotopic if and only if their Pontryagin manifolds are framed cobordant.

Theorem C. Every compact framed submanifold of codimension p in M is the Pontryagin manifold of some smooth map $M \to S^p$.

Together: **homotopy classes of maps $M \to S^p$ correspond bijectively to framed cobordism classes of codimension-p framed submanifolds of M.**

Lemma 2.1 (Milnor's Lemma 1). If v, v' are two positive bases of $T(S^p)_y$, then $(f^{-1}(y), f^*v)$ and $(f^{-1}(y), f^*v')$ are framed cobordant.

Proof. The positive bases of $T(S^p)_y$ correspond to matrices in $GL^+(p)$, which is path connected (Chapter 6, Lemma 1.1). Choose a smooth path v_t of positive bases from v to v', constant near $t = 0$ and $t = 1$. Then $X := f^{-1}(y) \times [0,1]$ with framing $u^i(x,t) := ((f^*v_t)^i(x), 0)$ is a framed cobordism. (The normal space of X in $M \times [0,1]$ at (x,t) is $TN_x^\perp \times \{0\}$.) □

We now often write "the framed manifold $f^{-1}(y)$" for the Pontryagin manifold with any positive basis.

Lemma 2.2 (Milnor's Lemma 2). If y is a regular value of f and z is sufficiently close to y, then $f^{-1}(z)$ is framed cobordant to $f^{-1}(y)$.

Proof. The critical values of f form a compact set (image of the closed set of critical points of the compact M), so there is $\varepsilon > 0$ such that every point within distance ε of y is a regular value. Let $\|z - y\| < \varepsilon$, $z \neq y$. Let $\theta \in (0, \pi)$ be the angle between y and z and R_α the rotation by angle α in the plane spanned by y, z, fixing its orthogonal complement. Let $\beta : [0,1] \to [0,1]$ be smooth, 0 near 0 and 1 near 1, and $r_t := R_{\beta(t)\theta}$. Then r_t is a smooth family of rotations of S^p with $r_t = \mathrm{id}$ near $t = 0$, $r_t = r_1$ near $t = 1$, $r_1(y) = z$, and each $r_t^{-1}(z) = R_{-\beta(t)\theta}(z)$ lies on the short great-circle arc from z to y, all of whose points are within distance $\|z - y\| < \varepsilon$ of y (chord length increases with angle), hence are regular values of f.

Define $F : M \times [0,1] \to S^p$, $F(x,t) := r_t(f(x))$. For each t, z is a regular value of $r_t \circ f$: its preimage is $f^{-1}(r_t^{-1}z)$, a set of regular points of f, and $d(r_t \circ f) = dr_t \circ df$ is onto. A fortiori z is a regular value of F, and of F restricted to $M \times \{0,1\}$. By Chapter 2, Lemma 4.1, $X := F^{-1}(z)$ is a compact manifold with boundary $f^{-1}(z) \times \{0\} \cup f^{-1}(y) \times \{1\}$. Near $t = 0$, $F(x,t) = f(x)$, so X is $f^{-1}(z) \times [0, \varepsilon_0)$ there and the framing F^*v is $(f^*v, 0)$; near $t = 1$, $F(x,t) = r_1 f(x)$, so X is $f^{-1}(y) \times (1 - \varepsilon_0, 1]$ with framing $((r_1 f)^*v, 0) = (f^*(dr_1^{-1}v), 0)$, where $dr_1^{-1}v$ is a positive basis at y (rotations preserve orientation). So (X, F^*v) is a framed cobordism from $f^{-1}(z)$ to $f^{-1}(y)$ with the framing $f^*(dr_1^{-1}v)$, which is framed cobordant to $f^{-1}(y)$ with any positive framing by Lemma 2.1. □

Lemma 2.3 (Milnor's Lemma 3). If f and g are smoothly homotopic and y is a regular value of both, then $f^{-1}(y)$ and $g^{-1}(y)$ are framed cobordant.

Proof. Reparametrizing the homotopy (Chapter 4, §1), choose $F : M \times [0,1] \to S^p$ with $F(x,t) = f(x)$ for $t \leq \varepsilon$ and $F(x,t) = g(x)$ for $t \geq 1 - \varepsilon$. By Brown's theorem choose a regular value z of F so close to y that, by Lemma 2.2, $f^{-1}(z)$ is framed cobordant to $f^{-1}(y)$ and $g^{-1}(z)$ to $g^{-1}(y)$. Because F is constant in t near the ends, $dF_{(x,t)} = (df_x, 0)$ for $t \leq \varepsilon$, so z regular for F

makes z regular for f, and likewise for g. Then $F^{-1}(z)$ with the framing F^*v is a framed cobordism from $f^{-1}(z)$ to $g^{-1}(z)$, exactly as in the previous proof. Chain the three cobordisms. $\square$

Proof of Theorem A. Independence of v is Lemma 2.1. For two regular values y, z: choose rotations r_t as above with $r_0 = \mathrm{id}$, $r_1(y) = z$ (any two points of the sphere are joined by a rotation path). Then $F(x,t) = r_t f(x)$ is a homotopy from f to $r_1 \circ f$, and z is a regular value of both ($(r_1 f)^{-1}(z) = f^{-1}(y)$). By Lemma 2.3, $f^{-1}(z)$ is framed cobordant to $(r_1 f)^{-1}(z) = f^{-1}(y)$ (with the framing $f^*(dr_1^{-1}v)$, positive, hence by Lemma 2.1 equivalent to any). $\square$

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3. The product neighbourhood theorem (Milnor pp. 45–46)

Theorem 3.1. Let (N, v) be a compact framed submanifold of codimension p in M (both without boundary). Then there is a diffeomorphism $g : N \times \mathbb{R}^p \to V$ onto an open neighbourhood V of N in M, with $g(x,0) = x$ and $dg_{(x,0)}(0, e_i) = v^i(x)$.

Milnor proves this for $M = \mathbb{R}^{n+p}$ by straight lines and says the general case uses geodesics. We can avoid geodesics with the tubular neighbourhood of Chapter 6.

Proof. Let $r : N_\varepsilon \to M$ be the smooth closest-point retraction of Chapter 6, Proposition 4.1, defined on the ε-neighbourhood of M in $\mathbb{R}^k$. Since N is compact, $C := \max_{x,i} \|v^i(x)\| < \infty$; let $\delta := \varepsilon/(pC)$ and U_δ the open δ-ball in $\mathbb{R}^p$. Define

$$g_0 : N \times U_\delta \to M, \qquad g_0(x;t) := r\Big(x + \sum_i t_i v^i(x)\Big),$$

well defined since the argument is within ε of $x \in M$. It is smooth. Its derivative at $(x,0)$: on TN_x it is $dr_x = \mathrm{id}$ (as r is the identity on M); on $\mathbb{R}^p$ it is $t \mapsto dr_x(\sum t_i v^i(x)) = \sum t_i v^i(x)$, because $v^i(x) \in TM_x$ and dr_x is the identity on TM_x. So $dg_{0,(x,0)} : TN_x \oplus \mathbb{R}^p \to TM_x = TN_x \oplus TN_x^\perp$ is an isomorphism. By the inverse function theorem g_0 is a diffeomorphism near each $(x,0)$. It is injective on $N \times U_{\delta'}$ for some $\delta' \le \delta$: otherwise there are pairs $(x_j, t_j) \neq (x_j', t_j')$ with $t_j, t_j' \to 0$ and equal images; passing to subsequences $x_j \to x_0$, $x_j' \to x_0'$ by compactness of N, continuity gives $x_0 = g_0(x_0, 0) = g_0(x_0', 0) = x_0'$, and for large j both pairs lie in a neighbourhood of $(x_0, 0)$ on which g_0 is injective, a contradiction. Thus g_0 maps $N \times U_{\delta'}$ diffeomorphically onto an open set $V \supseteq N$. Finally compose with the diffeomorphism $\mathbb{R}^p \to U_{\delta'}$ inverse to $u \mapsto u/(1 - \|u\|^2/\delta'^2)$ (a radial map with positive radial derivative, tending to infinity at the boundary), whose derivative at 0 is the identity. The resulting g has $g(x,0) = x$ and $dg_{(x,0)}(0, e_i) = v^i(x)$. $\square$

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4. Every framed submanifold is a Pontryagin manifold (Milnor pp. 46–47)

Proof of Theorem C. Let (N, v) be given, and $g : N \times \mathbb{R}^p \to V$ a product neighbourhood. Define the projection $\pi : V \to \mathbb{R}^p$, $\pi(g(x,u)) = u$. It is smooth, 0 is a regular value (in the coordinates g, π is the projection), $\pi^{-1}(0) = N$, and the framing $\pi^*(e_1, \dots, e_p)$ is v: the normal vector $w^i(x)$ with $d\pi_x(w^i) = e_i$ is $dg_{(x,0)}(0, e_i) = v^i(x)$, which is normal to N by hypothesis.

A map from $\mathbb{R}^p$ to the sphere. Fix a point $s_0 \in S^p$, let $h : S^p \setminus \{s_0\} \to \mathbb{R}^p$ be stereographic projection from s_0, and let $\lambda : \mathbb{R} \to [0, \infty)$ be smooth with $\lambda(t) > 0$ for $t < 1$, $\lambda(t) = 0$ for $t \geq 1$, and λ decreasing on $[0, 1)$; for instance $\lambda(t) = e^{-1/(1-t)}$ for $t < 1$. Define

$$\varphi(u) := h^{-1}(u/\lambda(\|u\|^2)) \text{ for } \|u\| < 1, \qquad \varphi(u) := s_0 \text{ for } \|u\| \geq 1.$$

On the open unit ball, φ is a diffeomorphism onto $S^p \setminus \{s_0\}$: the map $u \mapsto u/\lambda(\|u\|^2)$ is radial, with radial function $s \mapsto s/\lambda(s^2)$ strictly increasing from 0 to ∞ on $[0, 1)$ with positive derivative, so it is a diffeomorphism of the ball onto $\mathbb{R}^p$. Smoothness of φ across $\|u\| = 1$: in the stereographic chart h' from the antipode of s_0, which sends s_0 to 0 and is related to h by the inversion $h' = h/\|h\|^2$, we have $h'(\varphi(u)) = \lambda(\|u\|^2)\, u/\|u\|^2$ for $0 < \|u\| < 1$, and this extends by 0 to a smooth function on $\{u \neq 0\}$, because λ and all its derivatives vanish as $\|u\| \to 1$. So φ is smooth, and $\varphi(0)$ is a regular value with $\varphi^{-1}(\varphi(0)) = \{0\}$.

The map. Define $f : M \to S^p$ by $f := \varphi \circ \pi$ on V and $f := s_0$ on $M \setminus V$. The set $K := g(N \times \bar{B}_1) = \{x \in V : \|\pi(x)\| \leq 1\}$ is compact, hence closed in M, and $f \equiv s_0$ on the open set $M \setminus K$; on the open set V, f is smooth. So f is smooth on M. Its Pontryagin manifold at $\varphi(0)$, with the basis $(d\varphi_0 e_i)$ (positive for a suitable orientation of S^p; otherwise replace e_1 by $-e_1$ and v^1 by $-v^1$ throughout), is $\pi^{-1}(0) = N$ with framing $\pi^*(e_i) = v$. $\square$

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5. The Pontryagin manifold determines the homotopy class (Milnor pp. 47–49)

Lemma 5.1 (Milnor's Lemma 4). Let $f, g : M \to S^p$ have a common regular value y and *identical* Pontryagin manifolds there: $f^{-1}(y) = g^{-1}(y) =: N$ and $f^*v = g^*v$. Then f and g are smoothly homotopic.

Proof. **Step 1: the case $f = g$ on a neighbourhood V of N.** Let $h : S^p \setminus \{y\} \to \mathbb{R}^p$ be stereographic projection from y. For $x \notin N$, $f(x) \neq y$ and $g(x) \neq y$, so

$$H(x,t) := \begin{cases} f(x), & x \in V, \\ h^{-1}\big(t\, h(f(x)) + (1-t)\, h(g(x))\big), & x \in M \setminus N \end{cases}$$

is well defined (the two formulas agree on $V \setminus N$, where $f = g$), smooth, and is a homotopy from g to f.

Step 2: making f agree with g near N. Since $f^*v = g^*v$ means $df_x = dg_x$ on the normal spaces, and both vanish on TN_x, we have $df_x = dg_x$ for $x \in N$. Choose a product neighbourhood $g_N : N \times \mathbb{R}^p \to V$ (Theorem 3.1) so small that $f(V)$ and $g(V)$ avoid the antipode $\bar{y}$ of y (possible: $f^{-1}(\bar{y})$ is a compact set disjoint from N). Let $\bar{h} : S^p \setminus \{\bar{y}\} \to \mathbb{R}^p$ be stereographic projection from $\bar{y}$, normalized so that $\bar{h}(y) = 0$ and $d\bar{h}_y(v_i) = e_i$ (compose with a linear map). Put

$$F := \bar{h} \circ f \circ g_N, \qquad G := \bar{h} \circ g \circ g_N : N \times \mathbb{R}^p \to \mathbb{R}^p.$$

Then $F^{-1}(0) = G^{-1}(0) = N \times \{0\}$, and at $(x, 0)$ both dF and dG vanish on TN_x and send $(0, e_i) \mapsto d\bar{h}_y(df_x(w^i(x))) = d\bar{h}_y(v_i) = e_i$: both derivatives are the projection to $\mathbb{R}^p$.

A lower bound. By Taylor's theorem on the compact set $N \times \bar{B}_1$ (second derivatives bounded), there is c_1 with $\|F(x,u) - u\| \le c_1\|u\|^2$ for $\|u\| \le 1$, hence

$$F(x,u) \cdot u \;\ge\; \|u\|^2 - c_1\|u\|^3 \;>\; 0 \qquad \text{for } 0 < \|u\| < c := \min(1/c_1, 1),$$

and similarly for G (take the worse constant). So for $0 < \|u\| < c$, $F(x,u)$ and $G(x,u)$ lie in the same open half-space $\{w : w \cdot u > 0\}$, and no convex combination of them is 0.

The deformation. Let $\lambda : \mathbb{R}^p \to [0,1]$ be smooth, 1 for $\|u\| \le c/2$ and 0 for $\|u\| \ge c$. Put

$$F_t(x,u) := (1 - \lambda(u)\,t)F(x,u) + \lambda(u)\,t\,G(x,u).$$

Then $F_0 = F$; $F_t = F$ wherever $\|u\| \ge c$; $F_1 = G$ wherever $\|u\| \le c/2$; and F_t has no zeros with $u \ne 0$: for $\|u\| \ge c$ because F has none, for $0 < \|u\| < c$ because $F_t(x,u) \cdot u > 0$. Transport back: define $f_t := \bar{h}^{-1} \circ F_t \circ g_N^{-1}$ on V and $f_t := f$ outside $g_N(N \times \bar{B}_c)$. The two definitions agree on the overlap (where $F_t = F$), and $g_N(N \times \bar{B}_c)$ is compact, so f_t is a smooth homotopy from f to a map f_1 with $f_1^{-1}(y) = N$ and $f_1 = g$ on the neighbourhood $g_N(N \times B_{c/2})$ of N. By Step 1, $f_1 \simeq g$. Hence $f \simeq g$. □

Proof of Theorem B. If $f \simeq g$, their Pontryagin manifolds are framed cobordant by Lemma 2.3 (choose a common regular value, which exists as in Chapter 5). Conversely let (X, u) be a framed cobordism in $M \times [0,1]$ from $(f^{-1}(y), f^*v)$ to $(g^{-1}(y), g^*v)$. We build a homotopy $F : M \times [0,1] \to S^p$ whose Pontryagin manifold at y is exactly (X, u), by the construction of Theorem C applied to X:

Product neighbourhood of X. Run the proof of Theorem 3.1 with $M \times [0,1] \subseteq \mathbb{R}^{k+1}$ in place of M and X in place of N: the retraction is $r \times \mathrm{id}$, and $g_0(\xi; t) = (r \times \mathrm{id})(\xi + \sum t_i u^i(\xi))$ for $\xi \in X$. For ξ in the end regions $N \times [0, \varepsilon)$ and $N' \times (1-\varepsilon, 1]$ the framing vectors are horizontal, $u^i = (v^i, 0)$, so g_0 preserves the $[0,1]$-coordinate there and maps into $M \times [0,1]$; for ξ with $[0,1]$-coordinate in $[\varepsilon, 1-\varepsilon]$, a displacement of size less than $\varepsilon/2$ stays inside $M \times (0,1)$. So for small δ, g_0 is defined on $X \times U_\delta$ with values in $M \times [0,1]$; the derivative and

injectivity arguments are unchanged (X is compact), giving a diffeomorphism $g_X : X \times \mathbb{R}^p \to W$ onto an open neighbourhood W of X in $M \times [0,1]$, which restricts over $N \times \{0\}$ to a product neighbourhood of N in $M \times \{0\}$ with the framing $v = f^*v$, and similarly at $t = 1$.

The homotopy. Define $\Pi : W \to \mathbb{R}^p$ by $\Pi(g_X(\xi, u)) = u$ and $F := \varphi \circ \Pi$ on W, $F := s_0$ elsewhere, with φ as in Theorem C and s_0 chosen so that $\varphi(0) = y$ and $d\varphi_0(e_i) = v_i$ (compose φ with a rotation and, if needed, a reflection of $\mathbb{R}^p$). As in Theorem C, F is smooth, and its Pontryagin manifold at y is (X, u). The map $F_0 := F(\cdot, 0)$ has $F_0^{-1}(y) = N$ with framing $F_0^* v = v = f^*v$: the Pontryagin manifolds of F_0 and f at y are *identical*. By Lemma 5.1, $F_0 \simeq f$; likewise $F_1 \simeq g$; and F is a homotopy from F_0 to F_1. Therefore $f \simeq g$. □

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6. The Hopf theorem (Milnor p. 50)

Now take $p = m$: maps $f : M \to S^m$ from a compact connected oriented m-manifold. A framed submanifold of codimension m is a finite set of points x with a basis $w(x)$ of TM_x. Define $\operatorname{sgn}(x) = +1$ if $w(x)$ is a positive basis of TM_x and -1 otherwise.

Claim 6.1. For the Pontryagin manifold of f at y, $\sum_x \operatorname{sgn}(x) = \deg f$.

Proof. $w(x) = f^*v$ is the basis with $df_x(w^i) = v_i$, v positive at y. It is positive iff df_x preserves orientation, i.e. iff $\operatorname{sign} df_x = +1$. Sum over $f^{-1}(y)$. □

Claim 6.2. Two framed 0-manifolds in M with the same $\sum \operatorname{sgn}$ are framed cobordant.

Proof. Three moves. *(a) Changing the frame at a point.* Two bases of TM_x of the same sign are joined by a path in GL^+; the cobordism $\{x\} \times [0,1]$ with the moving frame realizes this (as in Lemma 2.1). *(b) Moving a point.* Given x, x', choose a smooth path γ in M from x to x', constant near its ends (M is connected). The arc $X = \{(\gamma(t), t)\}$ with a frame obtained by continuously extending $w(x)$ along the arc (the normal bundle of an arc is trivial: extend a frame step by step and orthonormalize) is a framed cobordism from $(x, w(x))$ to (x', w') for some frame w' at x', and sgn is preserved: along the arc, the basis (positive tangent of X, $w^1, \dots, w^m$) of $T(M \times [0,1])$ has constant orientation, the tangent points in the $+t$ direction at both ends here, and (∂_t, w) is positive iff w is positive for M. *(c) Cancelling a pair of opposite signs.* Given $x \neq x'$ with $\operatorname{sgn}(x) = -\operatorname{sgn}(x')$, choose an embedded arc X in $M \times [0,1)$ from $(x, 0)$ to $(x', 0)$, vertical near its ends (for $m \geq 2$ join x to x' by an embedded path γ in M and take $(\gamma(s), \varepsilon \sin \pi s)$ suitably reparametrized; for $m = 1$, $M = S^1$ and an arc of the circle serves). Extend $w(x)$ to a frame along X normal to X. At the far end the tangent of X points in the $-t$ direction (the arc returns to $M \times \{0\}$), so by the constancy of orientation of (tangent, frame) along X, the transported frame at x' has sign $-\operatorname{sgn}(x) = \operatorname{sgn}(x')$; by (a), adjust it near the end to equal

$w(x')$. Then X is a framed cobordism from $\{x, x'\}$ to the empty set. Reading it backwards creates a pair.

Given two framed 0-manifolds with the same sum d: cancel opposite pairs in each until each consists of $|d|$ points of the same sign $\operatorname{sgn} d$ (if $d = 0$, until empty); then move the points of one onto the points of the other by (b), and fix the frames by (a). □

Theorem 6.3 (Hopf). Let M be compact, connected, oriented, without boundary, of dimension m. Two smooth maps $M \to S^m$ are smoothly homotopic if and only if they have the same degree.

Proof. Homotopic maps have equal degrees (Chapter 5, Theorem B). Conversely, if $\deg f = \deg g$, their Pontryagin manifolds have the same $\sum \operatorname{sgn}$ (Claim 6.1), hence are framed cobordant (Claim 6.2), hence $f \simeq g$ (Theorem B). □

Together with Chapter 5, §5 (every integer is a degree: compose $S^m \to S^m$ of degree k from a degree-k map of S^1 by suspension, or use Theorem C on $|k|$ framed points), this says that the homotopy classes of maps $S^m \to S^m$ are in bijection with $\mathbb{Z}$ via the degree: in the notation of homotopy theory, $\pi_m(S^m) \cong \mathbb{Z}$. Problem 4 of Chapter 8 removes the word "smooth".

> **Two remarks without proof (Milnor p. 50).** If M is connected but non-orientable, two maps $M \to S^m$ are homotopic iff they have the same degree modulo 2: sliding a point around a loop that reverses orientation changes the sign of its frame, so only the parity of the number of points survives. And Pontryagin used framed 1- and 2-manifolds in S^m to show that there are exactly two homotopy classes of maps $S^{p+1} \to S^p$ and $S^{p+2} \to S^p$ for $p \geq 3$; beyond that the method becomes unwieldy, and homotopy theory took over. But the construction runs both ways, and it is the seed of Thom's cobordism theory.

An interactive look at the construction

The figure shows a map from the torus $T^2 = S^1 \times S^1$ (drawn as a square with opposite edges identified) to the circle S^1: a direction field on the torus,

$$\theta(x, y) = 2\pi(nx + my) + (\text{a smooth periodic wobble}),$$

where n, m are integers. Choose a target direction y_0 with the slider. The curves are the Pontryagin manifold $\theta^{-1}(y_0)$, and the little arrows along them are the framing: the normal direction in which θ increases. Change the target and the wobble: the curves move, split and rejoin, but always by a framed cobordism, and the pair of winding numbers of the family never changes. Change n or m and it does, because integers cannot be changed continuously. Theorem B says these winding numbers, with the framing, are all a map $T^2 \to S^1$ remembers up to homotopy.

(In the printed version this figure is a still image; the interactive version is in the web edition.)

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7. Application: skyrmions and hopfions

Skyrmions. In a thin magnetic film the magnetization is a unit vector field $n : \mathbb{R}^2 \to S^2$ that tends to a fixed direction n_∞ (say "up") far away. Adding a point at infinity to the plane, as in Chapter 1, makes n a smooth map $\bar{n} : S^2 \to S^2$, and it has a degree $Q = \deg \bar{n} \in \mathbb{Z}$, the *skyrmion number*. In the physics literature it is written

$$Q = \frac{1}{4\pi} \int_{\mathbb{R}^2} n \cdot \Big(\frac{\partial n}{\partial x} \times \frac{\partial n}{\partial y} \Big)\, dx\, dy,$$

the integral of the Jacobian determinant of n (with respect to the area form of S^2); that the integral of the Jacobian of a map between compact oriented surfaces equals the degree times the target's area follows from the change of variables formula, applied over the regular values (which have full measure by Sard) using the local inverses of Chapter 1. A skyrmion is a texture with $Q = \pm 1$: a region where the magnetization points *down* at one point and sweeps around all directions as one moves outward. By Hopf's theorem, textures with different Q cannot be deformed into each other; by Theorem B, the Pontryagin manifold, which for $p = m = 2$ is the finite set of points where n points down, counted with signs, is all that matters. The topological protection of Q is why skyrmions are pursued as bits for magnetic memory: a skyrmion cannot fade away by a small perturbation. They were first seen as a lattice in manganese silicide by neutron scattering in 2009 and imaged directly the year after.

Hopfions. In a three-dimensional sample, $n : \mathbb{R}^3 \to S^2$ with $n \to n_\infty$ at infinity is a map $\bar{n} : S^3 \to S^2$. Now $m = 3 > p = 2$: no degree, but a Pontryagin manifold, which is a framed *link* in S^3, the set of points where n points north, with a normal frame. The framed cobordism class of a link in S^3 carries an integer, the *Hopf invariant* (Chapter 8, Problem 14): the linking number of the preimages of two distinct regular values, equivalently the linking number of the north curve with the south curve, or the self-linking of the framed north curve. For the Hopf map $S^3 \to S^2$ the preimages are two circles linked once. A texture with nonzero Hopf invariant is a *hopfion*; it cannot be deformed to the uniform state, because the linking of two closed curves cannot change without the curves crossing, which would mean two different directions at the same point. Hopfions were predicted in field theory (Faddeev 1975), found numerically as knotted solitons (Battye and Sutcliffe 1998), and observed as "hopfion rings" in the chiral magnet FeGe by electron holography in 2023. The photograph in that paper is, quite literally, a Pontryagin manifold: a closed loop of points where the magnetization points one way, threading a skyrmion string.

Why the construction, and not just the degree. The degree answers "how many times does the map wrap?" and only makes sense when domain and target have the same dimension. The Pontryagin manifold answers "where does the map hit a chosen value, and how does it approach it?" and makes sense whenever $m \geq p$. For a texture in a three-dimensional magnet the second question is the one you can photograph.

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8. Summary of what was silently assumed, now made explicit

Where	Milnor says	What is actually needed
p. 42	"there is a unique vector $w^i(x) \in Tf^{-1}(y)^\perp_x$ that maps to v_i"	df_x restricted to the normal space is an isomorphism, by Chapter 2, Lemma 2.2, and dimension count
p. 43	"the space of positive bases is connected"	$GL^+(p)$ path connected, Chapter 6, Lemma 1.1
p. 44	"each $r_t^{-1}(z)$ lies on the great circle from y to z, and hence is a regular value"	the short arc stays within the chord distance of y; critical values form a compact set (Lemma 2.2)
p. 44	$F^{-1}(z)$ "provides a framed cobordism"	F is constant in t near the ends, so the framing is horizontal there and z regular for F implies regular for f, g (Lemmas 2.2, 2.3)
p. 46	"replace straight lines by geodesics in M"	the closest-point retraction of Chapter 6 does the job without geodesics (Theorem 3.1)
p. 47	"choose a smooth map $\varphi : \mathbb{R}^p \to S^p \ldots$"; "clearly f is smooth"	explicit φ with a flat cutoff, checked in the chart at s_0; $f \equiv s_0$ on the complement of a compact set (Theorem C)
p. 48	"by Taylor's theorem $\|F(x,u) - u\| \leq c_1 \|u\|^2$"	$dF_{(x,0)}$ is the projection because $f^* v = g^* v$; compactness of $N \times \bar{B}_1$ (Lemma 5.1)

Where	Milnor says	What is actually needed
p. 49	"an argument completely analogous to the proof of Theorem C constructs a homotopy"	the product neighbourhood of X in $M \times [0,1]$ stays inside $M \times [0,1]$ because the framing is horizontal near the ends (Theorem B)
p. 50	"it is not difficult to see that the framed cobordism class of the 0-manifold is determined by $\sum$ sgn"	the three moves of Claim 6.2, with the orientation bookkeeping along an arc

Checked by machine. Everything in this chapter has been verified in the Lean proof assistant: framed cobordism, the Pontryagin construction, Lemmas 1–4, Theorems A, B and C, the Product Neighborhood Theorem and the Hopf theorem. Not formalized: the closing remark that maps from a non-orientable manifold are classified by the degree modulo 2. See the verified edition, §8, and *Milnor, proof by proof*, Parts V–VII.